67. A lightning rod with sharp tip reduces damage risk. Which one of the following is the primary reason for it?

(a) It attracts lightning directly.

(b) The sharp tip reduces the local electric field.

(c) It enhances field and promotes continuous corona discharge.

(d) It repels charged clouds.

68. A periscope uses two plane mirrors inclined at 45°. If the tube of the periscope is filled with a complete transparent liquid of refractive index 1·5, then which one of the following is correct in respect of the image seen?

(a) The image will remain unchanged.

(b) The image will appear laterally shifted.

(c) The image will appear 1·5 times brighter.

(d) The image will appear 1·5 times darker.

69. It has been observed that a mirror always forms the image between the pole and the focus, irrespective of the location of the object. Which one of the following is correct in respect of the nature of the mirror?

(a) It is a plane mirror.

(b) It is a convex mirror.

(c) It is a concave mirror of small focal length.

(d) It is a concave mirror of large focal length.

70. What is the upward force acting on a skydiver of mass 60 kg who is falling at a uniform speed of 2 m/s?

(a) 0

(b) 60 N

(c) 588 N

(d) 708 N

71. With respect to electromagnetic waves, the human eye is sensitive to

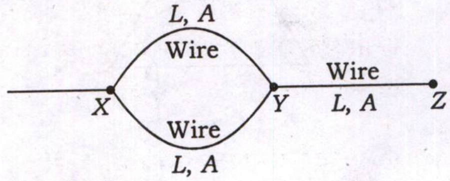
(a) both electric and magnetic fields

(b) electric field only

(c) magnetic field only

(d) infrared part of the electromagnetic spectrum

72. Three wires each of length L, cross-sectional area A and resistivity ρ are connected as shown in the figure. These are to be replaced by another wire of same resistivity such that the resistance between points X and Z does not change. If L_1 is the length and A_1 is the cross-sectional area of the new wire, then which one among the following is correct?



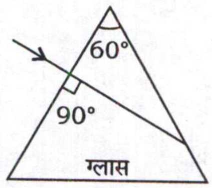
(a) $L_1 = 3L$ and $A_1 = 2A$

(b) $L_1 = L$ and $A_1 = A$

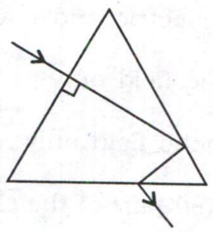
(c) $L_1 = 2L$ and $A_1 = 3A$

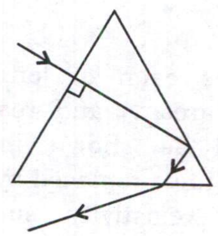
(d) $L_1 = 2L$ and $A_1 = 2A$

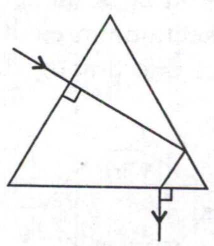
73. किसी समभुजी ग्लास प्रिज्म के किसी फलक पर आपतित किरण को चित्र में दर्शाया गया है :

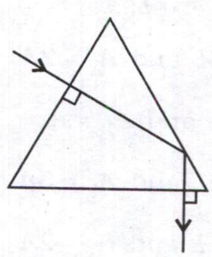


निम्नलिखित आरेखों में से कौन-सा एक ग्लास प्रिज्म से होकर गुजरने वाली किरण का सम्पूर्ण पथ सही दर्शाता है?

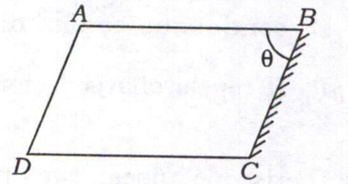
(a)


(b)


(c)


(d)


74. 1·5 अपवर्तनांक वाले एक काँच के स्लैब को चित्र में दर्शाए अनुसार एक समांतरषट्फलक *ABCD* के आकार में काटा जाता है। सतह *BC* को सम्पूर्ण परावर्तक बनाने के लिए पॉलिश किया जाता है। सतह *AB* पर अंदर से प्रकाश $\pi/3$ कोण पर आपतित है। कोण *ABC* (θ) का निम्नलिखित में से कौन-सा मान इस प्रकार सही है कि परावर्तित प्रकाश अपने पथ पर ही वापस लौटे?



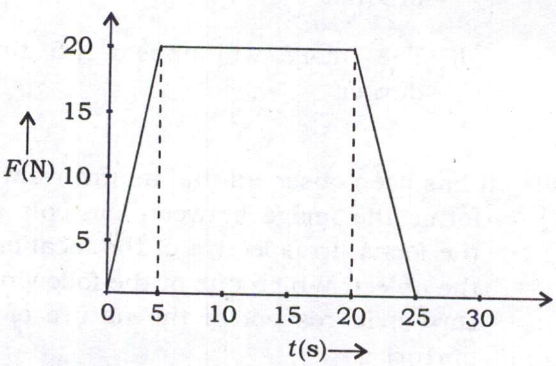
(a) $\pi/6$

(b) $\pi/3$

(c) $\pi/4$

(d) $5\pi/12$

75. 10 kg के एक गतिमान पिंड के लिए बल-समय (F - t) ग्राफ नीचे दर्शाया गया है :



पिंड की अंतिम चाल क्या है?

(a) 0

(b) 15 m/s

(c) 30 m/s

(d) 40 m/s